LARAVEL E-COMMERCE PLATFORM

MyShop Documentation

Complete technical documentation covering site architecture, features, database models, controllers, and security implementation

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Framework: Laravel 10+

Table of Contents

1. Project Overview
2. Site Structure
3. Database Models
4. Controllers
5. Middleware & Security
6. Routes
7. Problems Solved
8. Key Features
9. Views & Frontend
10. JavaScript Functions

PART 01

Project Overview

Project Information

Project Name	MyShop
Type	E-Commerce Platform
Framework	Laravel 10+
Database	SQLite
Frontend	Blade Templates + jQuery + Swiper.js
Auth System	Multi-guard (Admin + Customer)

Technology Stack

Backend

Laravel 10+, PHP 8.1+

Database

SQLite (database.sqlite)

Frontend

Icons

Bootstrap 5, jQuery, Swiper.js

Bootstrap Icons, Font Awesome

PART 02

Site Structure

Customer Pages (Frontend)

Home Page

Main landing page with featured products, countdown timers, and ads slider

`/, /show/home`

Products Listing

Product grid with search, filtering, and pagination

`/products`

Product Details

Detailed product view with ratings, color selection, and add-to-cart

`/products/{id}`

Cart

Shopping cart with quantity controls and totals

`/cart`

Checkout

Order placement form

`/checkout`

Customer Auth

Login, Register, and Social Login

`/customer/login, /customer/register`

Profile

Customer profile management and order history

`/profile, /show/orders`

Contact & About

Contact form and about us page

`/contact, /aboutus`

Admin Pages (Backend)

Admin Dashboard

Overview statistics and recent activity

`/admin/index`

Product Management

CRUD operations for products

`/admin/products`

Employee Management

Staff accounts with role-based access

Order Management

Order approval and rejection

`/admin/employees`

`/admin/orders`

Complaints

Customer complaint handling

`/admin/complaints`

Notifications

Admin notification center

`/admin/notifications`

Settings

Site configuration with 7 sections

`/admin/settings`

Videos

Video content management

`/admin/videos`

PART 03

Database Models

Core Models

Model	Table	Key Fields
`Product`	products	name, description, price, discount_percentage, stock, sku, img, category_id, is_new
`Order`	orders	customer_id, order_number, total_price, status, notes, expires_at, payment_method
`OrderItem`	order_items	order_id, product_id, quantity, price
`Cart`	carts	customer_id, product_id, quantity, color
`Customer`	customers	name, email, password, avatar, provider, provider_id
`Rating`	ratings	user_id, product_id, rating (1-5)
`Employee`	employees	name, email, password, avatar, status
`Role`	roles	name, permissions
`Complaint`	complaints	customer_id, subject, message, status

Product Model Key Methods

```
// Get discounted price (auto-calculates if within discount period)
$product->discounted_price;

// Check if product is currently on discount
$product->isOnDiscount(); // True if created within 8 days

// Get average rating
$product->averageRating();
```

Cart Model Color Mapping

The Cart model includes a `color_class` accessor that maps Arabic/English color names to CSS colors:

Supported colors: أسود, أحمر, أزرق, أخضر, أصفر, أبيض, وردي, بنفسجي, برتقالي, رمادي, بني, كحلي, تركواز, لافندر, زيتي, فضي, ذهبي, بيج, عنابي, كريمي, فيروزي, نيلي + English equivalents

PART 04

Controllers

Customer Controllers

Controller	Methods	Purpose
`ShowController`	home, products, details, about	Public-facing pages (home, products, product details)
`CartController`	index, add, update, remove, clear	Shopping cart AJAX operations
`CheckoutController`	index, store	Order placement
`CustomerAuthController`	showLoginForm, login, showRegisterForm, register, logout	Customer authentication
`SocialLoginController`	redirectTo*, handle*Callback	Google, Facebook, Twitter, Apple OAuth
`RatingController`	store	Product rating (1-5 stars, AJAX)
`OrderController`	myOrders, store, downloadInvoice, index, approve, reject	Order management
`ProfileController`	index, update	Customer profile

Admin Controllers

Controller	Methods	Purpose
`DashboardController`	index	Admin dashboard stats
`ProductController`	index, create, store, edit, update, destroy	Product CRUD (Laravel Resource)

Controller	Methods	Purpose
`EmployeeController`	index, create, store, edit, update, destroy, toggleStatus	Employee management
`AuthController`	login, register, logout, updateAvatar	Admin authentication
`SettingsController`	index, update, clearCache, optimize, toggleMaintenance	Site settings management
`AdminComplaintController`	index, show, destroy	Complaint handling
`AdminNotificationController`	index, markAsRead, markAllAsRead, destroy, clearAll	Notification management

PART 05

Middleware & Security

Custom Middleware

EnhancedAdminAuth

Session fingerprinting, IP whitelist, concurrent session prevention, activity timeout

BruteForceProtection

Login attempt limiting with exponential backoff

InputSanitization

SQL injection and XSS prevention for all inputs

SecurityHeaders

XSS, CSRF, clickjacking, MIME-type sniffing protection

CustomerAuth

Customer guard middleware

SetLocale

Multi-language support

EnhancedAdminAuth Features

- **Session Fingerprinting:** Validates user agent + Accept-Language + user ID
- **IP Whitelist:** Configurable allowed IPs (localhost exempt in dev)
- **Concurrent Session Check:** Prevents multiple simultaneous logins
- **Activity Timeout:** Auto-logout after configurable inactivity (default 30 min)
- **Security Event Logging:** Tracks suspicious activities

InputSanitization Features

- Recursive sanitization for nested arrays
- SQL injection pattern removal (UNION, SELECT, DROP, etc.)
- XSS pattern removal (script tags, javascript:, event handlers)
- High-severity pattern blocking with 403 response

- Security logging for suspicious inputs

PART 06

Routes

Route Groups

Group	Middleware	Prefix
Admin Auth	brute.force, input.sanitize	/admin/login, /admin/register
Admin Panel	enhanced.admin.auth	/admin/*
Customer Pages	set.locale	/ (home, products, cart, etc.)
Protected Customer	customer.auth	/profile, /cart, /checkout, /orders

Social Login Routes

Provider	Redirect	Callback
Google	/auth/google	/auth/google/callback
Facebook	/auth/facebook	/auth/facebook/callback
Twitter/X	/auth/twitter	/auth/twitter/callback
Apple	/auth/apple	/auth/apple/callback

API Routes

- `GET /products/search` - AJAX product search
- `POST /product/{id}/rate` - Submit product rating
- `PUT /cart/update/{id}` - Update cart quantity
- `DELETE /cart/remove/{id}` - Remove from cart
- `POST /notifications/read/{id}` - Mark notification read

PART 07

Problems Solved

Bug Fixes

Cart AJAX Buttons Not Working

Problem: Cart increment/decrement/delete buttons weren't responding to clicks.
Solution: Fixed selectors in cart.js - changed `.increase` to `.qty-btn.increase` and added proper event delegation on `#cart-items`.

Employee Images Not Displaying

Problem: Employee images showing broken icons.
Solution: Database column is `avatar` not `image`. Updated all references and storage paths.

Pagination Links Broken

Problem: Pagination on products page not working.
Solution: Typo in selector: `products-conatainer` → `products-container`.

Product Images Cropped

Problem: Product images getting cropped in cards.
Solution: Changed CSS from `object-fit: cover` to `object-fit: contain` so full image is visible with white space if needed.

UI Improvements

Modern Cart Page Design

Redesigned with glass morphism, gradient backgrounds, and smooth animations.

Product Cards Grid

Added overlay action buttons, discount badges, stock progress bars, and countdown timers.

Interactive Star Rating

Hover effects, click-to-rate, AJAX submission with JSON response.

PART 08

Key Features

E-Commerce Features

Product Discount System

Auto-discount for 8 days after creation. Product model calculates discounted_price automatically.

24 Arabic Color Options

Complete color mapping with tooltips. Supports Arabic and English color names.

Multi-language Support

Language switcher with locale middleware. Routes: /lang/{lang}

Social Login

Google, Facebook, Twitter/X, Apple OAuth integration with automatic account creation.

Admin Features

Settings Page (7 Sections)

General, Appearance, Security, Payment, Email, Advanced, Danger Zone

Employee Management

CRUD with avatar upload, status toggle, role assignment

Bulk Operations

Bulk delete products and other data cleanup

Maintenance Mode

Toggle via Settings or Artisan command

Security Features

Brute Force Protection

Login attempt limiting with exponential backoff

Input Sanitization

SQL injection and XSS prevention on all inputs

Session Security

Security Headers

Fingerprinting, IP whitelist, concurrent session prevention

X-Frame-Options, CSP, X-XSS-Protection, etc.

PART 09

Views & Frontend

View Structure

```
resources/views/
├── show/                # Customer frontend
│   ├── layouts/
│   │   └── navbar.blade.php    # Main navbar (light theme)
│   ├── home.blade.php          # Home page
│   ├── products/
│   │   ├── index.blade.php     # Products listing
│   │   ├── details.blade.php   # Product details
│   │   └── partials/
│   │       └── products.blade.php  # Product grid component
│   ├── auth/
│   │   ├── cart.blade.php      # Cart page
│   │   ├── login.blade.php     # Login with social buttons
│   │   └── register.blade.php  # Register with social buttons
│   ├── checkout.blade.php      # Checkout page
│   ├── profile.blade.php       # Customer profile
│   └── orders/
│       └── index.blade.php     # Order history
├── admin/               # Admin panel
│   ├── layouts/
│   │   └── sidebar.blade.php   # Admin sidebar
│   ├── login.blade.php         # Admin login
│   ├── register.blade.php      # Admin register
│   ├── index.blade.php         # Dashboard
│   ├── settings.blade.php      # Settings (7 sections)
│   ├── products/               # Product CRUD views
│   ├── employees/             # Employee management
│   ├── complaints/            # Complaints index & show
│   └── notifications/         # Notification center
```

Public Assets

```
public/
├── js/
│   ├── cart.js       # Cart AJAX (jQuery)
│   └── home.js       # Clock, countdown, dark mode, sliders
├── css/              # Stylesheets
└── storage/          # Uploaded files (images, avatars)
```

PART 10

JavaScript Functions

cart.js (jQuery)

Function	Event	Description
`updateSummary()`	Helper	Updates subtotal and total in cart summary
`increase`	Click	Increases quantity via AJAX PUT
`decrease`	Click	Decreases quantity via AJAX PUT
`delete-btn`	Click	Removes item from cart via AJAX DELETE
`clear-cart`	Click	Clears entire cart via AJAX POST
`updateNavbarCartCount()`	Helper	Updates cart badge count in navbar

home.js (Vanilla JS)

Function	Description
`updateClock()`	Digital clock updating every second
`initializeProductTimers()`	Countdown timers for discounted products
`showSlide()`	Ads slider navigation
`dark-mode-toggle`	Toggle dark theme class on body
`Swiper (homeSlider)`	Home page hero slider with autoplay

Rating System (Inline JS)

Frontend: Star rating with hover effects and click handlers submitting to `/product/{id}/rate`

Backend: Returns JSON: `{"success": true, "message": "تم إضافة تقييمك ⭐", "average": 4.5}`

DOM Update: Updates `.average-rating` element with new average

APPENDIX

Setup & Dependencies

Required Packages

```
composer require laravel/socialite  # For social login
```

Required Configuration

In `config/services.php`, add OAuth credentials:

```
'google' => [
    'client_id' => env('GOOGLE_CLIENT_ID'),
    'client_secret' => env('GOOGLE_CLIENT_SECRET'),
    'redirect' => env('APP_URL').'/auth/google/callback',
],
'facebook' => [...],
'twitter' => [...],
'apple' => [...],
```

Database Migrations

Important: After adding social login, run:

`php artisan migrate`

Storage Setup

```
php artisan storage:link  # Create storage link in public/
```

Key Files Reference

File	Purpose
`app/Http/Controllers/SocialLoginController.php`	OAuth handlers
`database/migrations/*provider*.php`	Customer provider columns

File	Purpose
`resources/views/show/auth/login.blade.php`	Social login buttons
`routes/web.php`	Social auth routes